

ANAND VIKRAM

Power Electronics Engineer

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CAREER OBJECTIVE

Dedicated and results-driven Power Electronics Engineer with a strong background in hardware architecture, PCB design, and system-level simulation, seeking a Senior/ Lead Engineering role to contribute to innovative power conversion solutions. Eager to leverage hands-on expertise in Ansys, PSIM and MATLAB/Simulink to design, test and optimize high-performance and efficient power systems. Focused on contributing to automotive, renewable energy, or industrial applications while driving product excellence and advancing technology.

SUMMARY:

- Research and development engineering professional in power electronics domain with 3+ years of work experience
- Experience in requirements identification and collection from customer end, requirements documents preparation, circuit designing & simulation, component selection, prototype development, Hardware-Software Integration Testing, traceability analysis, product validation and production planning
- Experience in preparation of requirements documents like Master list of Requirements (MLR) and product requirement specification (PRS) and management of BOM, product design report, Reliability reports, FMEA, Costing, and other project documents.
- Knowledge in Military Standards – MIL-STD-704F, MIL-PRF- 711G.
- Experience in electrical testing and development of various test jigs against the test/ product requirements.
- Hands on experience in using instruments like Digital Oscilloscope, current probes, data logger, insulation and dielectric tester, power analyzer, thermal cameras.
- Gained experience in cost analysis, value engineering, material and component selection and procurement.
- Experience in using IHS Tool software database for BOM Preparation

WORK EXPERIENCE

SFO Technologies PVT LTD.

2022 – NOW

Role : Power Electronics Design Engineer

- Actively involved in end-to-end hardware development, including system architecture, circuit design, and component selection for power electronics applications.
- Responsibilities include designing and testing prototypes, performing simulations using PSIM and MATLAB/Simulink, and ensuring performance and reliability through validation.
- Regularly engage with customers to understand technical requirements, provide design support, and ensure project alignment.
- Take ownership of both design execution and product quality, contributing to cross-functional collaboration and successful project delivery.

Internship at KSEB In Distribution Sector & Substation

2019 – 2020

- Designing Electrical systems and their component, based on technical requirements and industry standards
- Diagnosing and solving electrical problems
- Installation, calibration, and upgradation of electrical systems
- Repair and replacement of faulty electrical components
- Assisting in the preparation of user manuals and operating instructions.
- Conducting inspections to ensure that products or systems are in a safe and functional condition

PROJECTS DONE

Development of High Frequency Stabilizer – 1.3 kVA

High frequency stabilizer is to stabilize or regulate the input voltage at 230V for TV and Home Theatres in India. The system has wide input range from 90V to 300V AC and frequency range from 47Hz to 53Hz, with an efficiency greater than 95% at full load. So, it act as a protection device and protect the appliances from voltage surge.

- Project management & Coordination of team and stakeholders internal and external to meet targets.
- Discussing with customers and collecting product requirements
- Development of hardware architecture and analysis on single stage (AC-AC) and double stage (AC-DC-AC) conversion
- Design and simulation of various converters such as PFC, Flyback and Full bridge inverter
- Selection of switching devices such as MOSFETs, IGBTs and various components based on loss calculations
- Preparation of Board Level Schematics, BOM and various design documents for PDR and CDR
- Component Selection and Cost Estimation and minimization.
- Support for PCB Layout.
- Functional Testing and Validation of the circuit
- Board bring up, troubleshooting and hardware – software integration.

10 kW and 5 kW Regulated Transformer Rectifier Units

A 300/150A Line Replaceable Unit (LRU) that converts electrical power in aircraft from nominal 115/200 volts alternating current (AC), 400 Hz, three phase aircraft electrical systems power to nominal 28 volt regulated direct current (DC) power which are required for operating aircraft DC loads.

- Discussing with customers and collecting product requirements.
- Preparation of requirements documents like Master List of Requirements (MLR) and Product Requirement Specification (PRS)
- Development of system level and hardware architecture to meet safety critical requirements and airborne constraints as per the MIL standards.
- Design and simulation of 28V/300A and 28V/150A Phase shifted full bridge converters (PSFB)
- Selection of switching devices such as MOSFETs, IGBTs and various components based on loss calculations
- Preparation of Board Level Schematics, BOM and various design documents for PDR and CDR
- Support for PCB Layout
- Cost Minimization and Procurement Activities
- Analysis and simulation of analog protection circuits using Softwares like TINA-TI, LTSpice
- Board Bring up, Troubleshooting and Hardware – Software Integrated testing activities

50 kW RECTIFIER UNIT

Designed with five numbers of identical 10kW units stacked in parallel. The three phase input voltage may vary from 207VAC to 253VAC (Phase/Neutral). The output of each 10KW will be 310VDC, 33A. Each 10KW module is electrically connected in parallel and delivers 310V, 162A to a common DC bus.

- Development of system level and hardware architecture to meet safety critical requirements as per the MIL standards.
- Selection of switching devices such as MOSFETs, IGBTs and various components based on loss calculations
- Preparation of Board Level Schematics, BOM and various design documents for PDR and CDR
- Functional Testing and Validation of the circuit
- Board Bring up, Troubleshooting and Hardware – Software Integrated testing activities

ACADEMIC PROJECTS DONE :

1. Multiple Output Transformerless DCDC Converter
2. High Gain DC-DC Converter for Renewable Energy Application
3. Automatic Smart Car Parking System Using IOT

SKILL

- Ansys Tools – Maxwell, HFSS, Co- Simulation
- Embedded based Tools – Mentor Xpedition, MATLAB SIMULINK, TINA-TI, LTSPICE, PTC Windchill, PSIM
- Mechanical Tools – DraftSight
- Programming Tools– Basic C
- MS Office – Word, Excel, Presentation & Visio

CO- CURRICULAR ACTIVITIES

- Published an IJAREEIE paper on 'High Gain DC-DC Converter for Renewable Energy Application' in Vol 10, Issue 12, December 2021.
- Attended IEEE INDICON Program held at Cochin University of Science and Technology in 2022.

EDUCATION

M.TECH., POWER ELECTRONICS

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY, KERALA

CGPA : 8.22

2020 - 2022

B. TECH., ELECTRICAL & ELECTRONICS ENGINEERING

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY, KERALA.

CGPA : 7.46

2015 - 2019

PERSONAL DETAILS

Languages Known : English (Professional), Malayalam(Native) , Hindi (Average)

Date Of Birth : 26th February 1997

Nationality : Indian

DECLARATION

I hereby declare that the above-mentioned information is correct to best of my knowledge and belief. I bear the responsibility for the correctness of the above-mentioned particulars.

ANAND VIKRAM

Place: Kochi